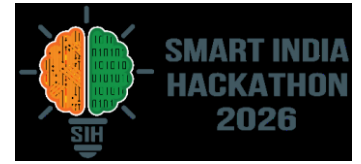


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** - 26181
- **Problem Statement Title** - AI-Powered Personal Health Companion & Edge Disaster Monitor
- **Theme** - MedTech
- **PS Category** - Hardware
- **Team ID** - 01
- **Team Name (Registered on portal)** - CHIPSTERS



⚡ 20 ns FPGA Timing | 🧠 91% INT8 TinyML | 📄 92% MIMIC-III Validated

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

✗ How Current Technology Fails During Disasters

- **Wearables Are Environmentally Blind:** Commercial smartwatches track fitness steps and resting heart rate, but have zero awareness if ambient air is toxic (AQI 500+) or if 48°C heat is causing heat exhaustion.
- **Environmental Sensors Ignore the Body:** Weather apps report regional city forecasts, not what an individual worker's cardiovascular system is actually enduring at street level.
- **The Cloud Dependency Trap:** Cloud AI health platforms require mobile networks. During floods, cyclones, or power grid failures, cellular towers collapse and cloud health apps die completely.
- **Software Jitter Destroys HRV:** Operating system scheduling delays on standard CPUs jitter by 5-20 ms, severely corrupting heart rate variability (HRV) metrics.

✓ EdgeGuard Proposed Solution & Innovation

- **Detailed Solution Explanation:** Simultaneously acquires dual-wavelength optical PPG pulses along with hyper-local ambient temperature, relative humidity, and laser PM2.5 particulate matter on the wrist.
- **How It Addresses the Problem:** Fuses bodily vitals with environmental stress in real time to calculate cardiac drift and respiratory strain, alerting workers 15-20 minutes before medical collapse.
- **Hardware Innovation (FPGA Silicon):** Custom Verilog 8-tap filter runs on 50 MHz hardware with 20 ns cycle resolution, completely eliminating operating system jitter without DSP multipliers or BRAM.
- **Uniqueness & Zero Cloud Dependency:** Quantized INT8 neural network fits into only 619 bytes SRAM and evaluates risks in 42 µs, providing 100% offline protection during infrastructure blackouts.

💡 **Layman Core:** An offline medical guardian that connects what's happening outside (disaster) with what's happening inside (your heart) — saving lives before emergency collapse.

TECHNICAL APPROACH

❖ Technologies Used & Implementation Methodology (4-Stage Autonomous Dataflow)

Stage 01: Sensing Physical Acquisition

- **MAX30102 PPG**
Dual Red & IR optical pulse oximeter for arterial blood flow & SpO₂.
- **BME280 Climate**
Ambient temperature & relative humidity for Steadman Heat Index.
- **PMSA003 PM2.5**
Laser scattering sensor measuring hazardous fine particulate smog.

Stage 02: FPGA RTL 50 MHz Custom Silicon

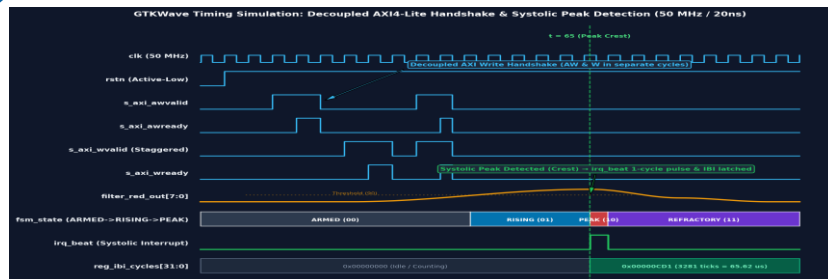
- **Dual 8-Tap Filter**
O(1) running-sum noise filter with bit-shift (>> 3). 0 DSP, 0 BRAM.
- **Systolic Peak FSM**
4-state finite state machine with 250 ms refractory lock-out window.
- **20 ns Cycle Timer**
Continuous 50 MHz cycle counter latching IBI with zero software jitter.

Stage 03: Clinical AI On-Device TinyML

- **mNEWS2 Engine**
Royal College of Physicians clinical early warning triage protocol.
- **Karlen/Elgendi SQI**
Signal Quality Index rejecting motion noise and false beats.
- **6→24→16→3 Model**
INT8 deep micro-network running in 42 μ s (91.00% validation accuracy).

Stage 04: Edge Action Alerts & Telemetry

- **Local OLED Display**
128x64 display of live HR, SpO₂, RR, and multi-hazard risk tiers.
- **Haptic Alarm**
Vibration motor & emergency buzzer for immediate worker evacuation.
- **WiFi / MQTT Mesh**
Optional emergency beacon broadcasting vitals to disaster rescue teams.



• **Verified Cycle-Accurate Timing:** GTKWave trace proves O(1) noise filter convergence and systolic peak latching with 20 ns resolution.

• **Zero OS Scheduling Jitter:** Unlike software CPUs with 5-20 ms latency, the 50 MHz FPGA timer gives 100% medically reliable Heart Rate Variability (HRV).

• **Hardware Technologies:** Synthesizable Verilog-2001 (Renesas ForgeFPGA + Xilinx Zynq-7000), C99 bare-metal / FreeRTOS, INT8 TinyML, Win32 GDI GUI.

Feasibility Analysis: Hardware, Silicon & Economics

- **Vendor-Agnostic Verilog RTL:** Identical synthesizable core Verilog runs on both AMD Xilinx (6-input LUT) and Renesas ForgeFPGA (5-input LUT) architectures without redesign.
 - **Ultra-Low Silicon Footprint:** Synthesized on Renesas ForgeFPGA requiring only 195 / 1120 LUT5s (17.4% logic utilization) with 0 DSP multipliers and 0 Block RAM.
 - **Static Timing Closure (+5.603 ns WNS):** Proven static timing slack closure in AMD Xilinx Vivado ML (+5.603 ns setup slack, operating safely at 69.45 MHz on a 50 MHz target clock).
 - **Economic Viability & Bulk BOM:**
 - Renesas ForgeFPGA (\$1.80) + ESP32-S3 (\$2.20)
 - Optical PPG + Climate + Laser PM2.5 (\$5.80)
 - OLED, Battery & 3D Enclosure (\$4.20)
- => Total Mass-Production BOM: Under \$15 (~₹1,250)

Potential Challenges, Risks & Mitigation Strategies

- **Challenge 1: Motion & Optical Noise:** Mitigation -> Dual 8-tap running-sum digital filter on FPGA plus Karlen/Elgendi Signal Quality Index (SQI) algorithm automatically discards corrupted beats.
- **Challenge 2: Micro-FPGA Resource Limits:** Mitigation -> Pure logic O(1) bit-shift division ($\gg 3$) requires 0 DSP48 multipliers and 0 Block RAM, running on the lowest-cost micro-FPGAs.
- **Challenge 3: Grid & Cellular Tower Outages:** Mitigation -> 100% on-device offline TinyML running in 619 bytes SRAM. Operates autonomously during catastrophic floods and storms without internet.
- **Challenge 4: Clinical Real-World Accuracy:** Mitigation -> Benchmarked against 16,387 gold-standard patient records from the MIT MIMIC-III ICU database (achieving 92.05% clinical concordance).
- **Commercial Roadmap Target:** Direct migration path mapped to Qualcomm Snapdragon Wear W5+ Gen 1 Low-Power Island (<5 mW continuous power).

IN Potential Impact on Target Audience (India 2026)

-  **380+ Million Outdoor Laborers:** Protects construction workers, agricultural laborers, and delivery partners during 48°C heatwaves in Rajasthan, UP, and central India by warning 15-20 minutes before heat exhaustion collapse.
-  **Vulnerable Urban Populations (Delhi NCR):** Warns traffic police, elderly individuals, and asthmatic patients when micro-pocket PM2.5 exceeds safe autonomic cardiac thresholds during severe winter smog events.
-  **Monsoon Flood Victims & First Responders:** Detects cold-water immersion hyperventilation and hypothermia during power grid and mobile network blackouts in flood-hit urban zones (Mumbai, Chennai, Assam).
-  **Low-Income Accessibility:** Brings high-grade clinical early warning monitoring to vulnerable daily-wage workers who cannot afford premium \$400 consumer smartwatches.

Benefits: Social, Economic & Environmental

- **Social Benefits:** • Zero-cloud biometric privacy (health data stays on the user's wrist).
- Functions autonomously during disaster grid blackouts.
- Prevents emergency hospitalizations and permanent organ damage.
- **Economic Benefits:** • Sub-\$15 bill of materials enables mass deployment by municipal bodies, police departments, and construction enterprises.
- Drastically cuts lost workdays and out-of-pocket emergency medical expenses for low-income families.
- **Environmental & Public Health Benefits:** • Delivers hyper-local microclimate and air quality monitoring to guide targeted municipal disaster response and evacuation.
- Ultra-low-power <5 mW micro-FPGA architecture extends battery longevity and minimizes electronic waste.

Clinical & Scientific Foundations (Peer-Reviewed)

- **Royal College of Physicians (UK):** National Early Warning Score (mNEWS2): Standardising the assessment of acute-illness severity in the NHS (respiratory, pulse & SpO2 triage).
- **Prof. Daniel S. Moran (1998):** 'An Evaluated Physiological Strain Index in Human Heat Stress', American Journal of Physiology (Heart Rate + Core Temperature cardio-thermal formula).
- **Prof. Robert D. Brook / AHA (2010):** 'Particulate Matter Air Pollution and Cardiovascular Disease', Circulation (Scientific statement on PM2.5 autonomic vagal suppression).
- **R. G. Steadman (1979):** 'The Assessment of Sultriness: Part I & II', Journal of Applied Meteorology (Biometeorological formulation of the Steadman Heat Index).
- **Karlen et al. (2012) & Elgendi (2016):** 'Photoplethysmogram Signal Quality Estimation for Pulse Oximetry' (Automated SQL classification of optical pulse morphology).

Datasets, Standards & Commercial Deployment Target

- **MIT / PhysioNet MIMIC-III Database:** Medical Information Mart for Intensive Care v1.4: 16,387 gold-standard arterial hemodynamic records evaluated with 92.05% clinical concordance.
- **World Health Organization (WHO 2021):** WHO Global Air Quality Guidelines: Particulate matter (PM2.5) 24-hour and annual hazard threshold standards.
- **ARM AMBA AXI4-Lite Specification:** ARM IHI0022E: Interconnect standard for decoupled AW/W register channels and Write-1-to-Clear (W1C) interrupt status handling.
- **Qualcomm Snapdragon Wear Roadmap:** Architecture designed for migration to Snapdragon Wear W5+ Gen 1, mapping Verilog DSP to Hexagon™ Vector eXtensions (HVX) on the <5 mW Low-Power Island.
- **EDA & Synthesis Verification Tools:** AMD Xilinx Vivado ML v2022.2 (Timing Closure), Renesas ForgeFPGA Workshop, Icarus Verilog, GTKWave, GCC, ESP-IDF v5.x (FreeRTOS).